

7 Recommendations for Stretching Prescription

If the goal of the stretching session is to induce a plastic increase in range of motion (ROM), then the stretching should not be performed as part of a warm-up or a cooldown. Stretching during these periods has different objectives and the possibility of negative performance implications for performance if the stretching is prolonged and performed in isolation before an activity (see Chapter 8). Details about stretching as part of the warm-up will be provided in a following section.

Stretching Duration

Almost any duration of stretching can improve ROM. In almost every study, you read, the very act of measuring a joint ROM will improve elastic flexibility. In one of our studies, we found we needed to pre-test individuals at least three times to ensure the ROM did not increase as a result only of the testing (33). Thus, a single stretch of less or equal to five seconds may improve ROM (65). That is why every stretching study needs a control condition or group! Whereas Roberts and Wilson (65) showed that nine stretches of five seconds provided similar increases in passive ROM as three stretches of 15 seconds, the 15 second stretches had a significantly better effect on active ROM than the 5 second duration stretches. Earlier studies by Bandy and Irion (4) suggested that 30–60 seconds of static stretching was more effective than 15 seconds to increase passive ROM. A static stretch training study had participants train three times a week for five weeks and found significant improvements in ROM with five seconds of stretch training but 15 seconds of stretching provided greater improvements (65). Other researchers also support using greater than 30 seconds of static stretching to achieve the greatest ROM (17, 27). In a later study by Bandy et al. (5) they indicated that one 30 second static stretch per day increased hamstrings ROM, but there were no differences with increased repetitions or frequency. In an animal study using rabbits (75), it was found that the greatest length changes in the MTU occurred within the first four stretches. In humans, two 30-second static stretches were needed to significantly decrease musculotendinous stiffness of the plantar flexors (68). There were no further decreases with 3–4 stretches of

the same duration. While we are obviously not the same as rabbits, the findings with these warm-blooded mammalian cousins in association with some of the human studies tend to provide the same message: there is no need for an exaggerated duration of stretching to obtain acute optimal increases in ROM. Thomas et al. (76) in their meta-analysis recommended a minimum duration of five minutes per week for each muscle group. Once again referring back to the Konrad et al.'s meta-analysis (42), they reported no stretch training ROM differences based on stretching volume or intensity.

Stretching Intensity

Many studies utilize static stretches that elongate the musculotendinous unit or joint to the point of discomfort (pain) or near the point of discomfort (6, 7, 8, 9, 10). Mechanically, this stretch point would be referred to as the elastic limit. The elastic limit is the minimum amount of stress placed on a tissue to elicit permanent strain. Exceeding the elastic limit will result in the tissue not returning to its original length after the stretch (1). At this point, musculotendinous strains or ligamentous sprain injuries could occur. Alter (1) expands on the strength training overload principle (28, 29, 30), transferring the concept to flexibility training as the overstretching principle. The overstretching principle is described as “when the body is regularly stimulated by an increasingly intense stretching programme beyond the homeostatic level, it will respond with an increased ability to stretch” (p. 145). The question arises, is it necessary to reach the elastic limit or point of discomfort to achieve plastic or semi-permanent changes in flexibility?

A number of acute studies have shown that submaximal intensity stretches provide similar ROM benefits as near maximal point of discomfort stretches (40, 41, 53, 85). We had subjects stretch at 100 percent, 75 percent, and 50 percent of the point of discomfort but did not find any significant difference in the flexibility test (stoop and reach). All conditions improved by approximately 12 percent (9). In contrast, another study reported that static stretching at 85–100 percent of maximum stretch intensity provided greater ROM than stretching at 60 percent (80). Other studies have compared high force-short duration to low force-long duration stretching and report that high force stretches emphasize elastic tissue deformation that shortly returns to its original length, whereas low force, prolonged stretching enhances plastic or semi-permanent changes in tissue length (48, 73, 81, 82). Apostopoulos (2) recommended stretching at 30–40 percent of perceived exertion. As a reminder from the last chapter, the Konrad et al. meta-analysis (42) revealed no ROM differences based on chronic stretch training volume, intensity, or weekly frequency. Stretching to the point of pain could be counterproductive. A typical response to pain or discomfort or distress is to adopt a stiffening strategy (16): that is to contract both the agonist and antagonist muscles in order to protect them from possible physical insults. Hence, while the individual is trying to lengthen the muscle, the central nervous system is trying to

shorten the muscle. So, there is no need for masochism when stretching, pain is not a necessity. Furthermore, stretching to the point of discomfort or elastic limit for a recently injured or fatigued tissue could strain (muscles and tendons) or sprain (ligaments) the tissue. Thus, during rehabilitation or after an intense, fatiguing, training session or match, high intensity stretching should not be pursued. Another point to consider is that pain is highly subjective to the individual. Hence, telling a person with a high pain threshold to stretch to the point of discomfort will place much greater stress on the tissues than for someone who has a lower tolerance to pain. It is much safer and reportedly equally effective to stretch below pain tolerance!

Continuous stretching (no rest periods) permits higher stretching intensities (31). However as mentioned, higher stretching intensities may not provide any extra ROM benefits. In fact, one study reported that intermittent stretching reduced peak torque (i.e., strength) more than continuous stretching, which were strongly associated with a depression in central drive (77).

Optimal Time of Day to Stretch

When is the best time of day to stretch? There are diurnal (time of day) variations in performance. Typically, most people perform maximal strength and power activities better in the afternoon compared to the morning. Dynamic stretching has been shown to counteract the lower vertical jump heights often found in the morning (18). Dr. Stuart McGill, a highly respected biomechanist (with one of the best moustaches around) from the University of Waterloo suggests that extensive stretching in the morning is counterproductive. Vertebral discs are infused with fluid known as the nucleus pulposa. After a night of being horizontal during sleep, the disc's nucleus pulposa become more hydrated. The gravity-induced fluid loss owing to a day of upright posture is replaced. This modulation in fluid alters the stresses on the disc throughout the day. Specifically, stress is highest following bedrest then diminishes over the subsequent few hours. Stretching the lower back in the morning with the discs expanded can increase the chances for disc herniation. Passive tissues of the back can be injured by bending over to pick up a pencil especially if the individual is unstable. Thus, attempting to touch the toes whether in the morning or any other time of the day can lead to injury. In fact, McGill proposes that a more flexible back actually increases the risk of future back problems. He states there is trade-off between mobility and stability and that balance is specific to the individual based on prior injuries, age, training objectives and other factors (55).

Stretching Frequency (Days/week)

Should we stretch every day or alternate days? The Konrad et al. meta-analysis (42) did not find any significant differences in ROM associated with the weekly frequency. Perhaps examining and contrasting with the strength training literature would be helpful on this issue. Resistance training the same

muscle on subsequent days is not advised since the muscle needs time to recover. If the intensity is sufficient, overload resistance stress can degrade muscle protein and activate a number of receptors and pathways for increased protein synthesis. This overcompensation to the stress can result in increased muscle strength and hypertrophy (43, 44, 45). However, without a sufficient recovery period (typically 48–72 hours), the muscle pathways continue to emphasize protein degradation rather than the synthesis of protein resulting in decreased strength and atrophy. For the most extreme example, contemplate the muscle hypertrophy of a bodybuilder to the muscle atrophy of a war-time concentration camp prisoner. Both individuals receive an overload stress on the muscle, but the bodybuilder builds appropriate rest periods (and of course proper nutrition) into the programme whereas the prisoner works hard every day without the chance for overcompensation. The difference between resistance training and stretching is the lower overload intensity and lack of protein degradation with stretching. Most stretching programmes would not lead to similar protein degradation or depletion of muscle glycogen stores as resistance training and thus there would not be a need for a prolonged recovery period. There is ample evidence of individuals who stretch every day providing significant improvements in flexibility without any negative consequences. It is possible that extreme flexibility programmes as seen with some gymnasts, dancers and other similar athletes may damage muscle and connective tissue and might be more effective with alternate day stretching. However, this type of comparative research has not been conducted. Flexibility training studies have successfully improved ROM with two (46), three (23), four (86), five (12), and seven (47, 52, 61, 67, 84) days/week flexibility training programmes. As can be seen from the number of citations, daily stretching has been commonly used in the literature. Thus, for the vast majority of people who stretch, daily stretching should provide substantial improvements in flexibility. As mentioned previously, stretching only 1 day / week can sustain previously attained flexibility training gains (79). Since no extensive direct comparison studies (i.e., one study that directly compared one vs. three vs. five vs. seven days of stretching per week) have been conducted, we cannot state whether daily stretching actually provides better flexibility improvements than five days/week, three days/week or other frequencies. However, the Thomas et al. (76) meta-analysis comparing the results of studies using differing stretching frequencies suggested that stretching at least five days per week (with a minimum five minutes per muscle group per week) provided the most beneficial increases in ROM. Furthermore, while Alter (1) suggests that stretching once a day would maintain flexibility and “empirical evidence suggests that stretching at least twice a day is preferable” (p. 154), there is scant evidence regarding whether stretching multiple times per day substantially enhances the improvement in flexibility.

Pre- versus post-workout stretching

Should we stretch prior to or after a workout? There is an entire section later in the text on the effects of pre-activity stretching during a warm-up, so we will leave that discussion till then. A common practice is to perform stretching exercises after a workout with the rationale being that the muscles and connective tissue are warm and the viscosity is low. This decreased viscosity is certainly an advantage for achieving greater muscle and tendon lengths, but dependent upon the prior activity, the muscles may be fatigued as well. Hence, attempting to stretch musculotendinous tissue to the maximum point of discomfort in order to achieve greater extensibility could lead to tissue strains if the tensile strength of the tissue is compromised by fatigue. Therefore, post-exercise stretching especially if fatigue is induced should involve low to moderate intensity stretching, so as not to overload the tension on the muscles and tendons. This stretching period should be more in the relaxation mode than to increase musculotendinous extensibility!

Stretching for Relaxation

Static stretching can physiologically relax an individual. Static stretching (five repetitions x 1 minute each of triceps surae) changed the predominance of the autonomic nervous system to a greater parasympathetic neural influence during the stretch and continued for four minutes following the stretch (38). This effect can last substantially longer as another study found the greater parasympathetic influence returning to pre-stretch levels 30 minutes after stretching (three stretches x 30 seconds each) (25). Relaxation can be an important aspect of exercise recovery. High stress whether it be physiological or psychological can increase cortisol (78). Cortisol, a catabolic hormone, increases protein degradation, increases the metabolism of protein, fat, and carbohydrates and suppresses the immune system (32). With recovery, we basically want the opposite actions to occur. Thus, while high intensity stretching is not recommended after a workout with the objective of increasing ROM, stretching in order to relax would be a strong recommendation after taxing physical activity. In addition to stretching, a focus on breathing patterns can affect parasympathetic activation. Yoga, of course has developed and focused on a variety of ventilatory strategies for centuries.

Ventilatory Effects on Stretching

A growing body of evidence supports the belief that yoga benefits physical and mental health via down-regulation of the hypothalamic-pituitary-adrenal axis and the sympathetic nervous system (3). With yoga, the emphasis is often on deep, rhythmic, consistent, diaphragmatic and nasal breathing to relax the individual (21). In a more relaxed state, it is believed that the individual will be able to achieve greater ROM. Unpublished data from our lab showed that

unilateral nasal breathing could affect heart rate (Kliger et al.) While walking on a treadmill for 10 minutes, with unilateral left nostril breathing, there were significant increases in heart rate, systolic and diastolic blood pressures. Post-treadmill walking, flexibility was tested. A significant decrease in flexibility was found. In contrast, greater flexibility was achieved, with decreased heart rate and blood pressure during walking after unilateral right nasal breathing. These results correspond with Yoga Tradition. Energy flow through “ida” (during left nasal breathing practice) is supposed to be ‘heat dissipating (cooling)’ whereas energy flow through “pingala” (during right nasal breathing practice) is ‘heat generating’. Hence, there may be a nostril laterality affecting the autonomous nervous system differentially (22), with left nostril breathing emphasizing the parasympathetic influence.

Similar to breathing or ventilatory patterns when resistance training, the breathing pattern can have an effect on stretching especially when performing a trunk forward flexion action. Expiring during trunk forward flexion is typically recommended as a full inspiration tends to contract the erector spinae muscles (15, 64) and expand the thoracic cage (ribs), which would detract from attempting to fully flex (1). In contrast, Hamilton and colleagues (34) found greater ROM and lower electromyography (EMG) activity of the rectus abdominus, external obliques, lower abdominal stabilizers, and lower erector spinae of women with breathing techniques that emphasized larger inhalations. There was no effect of different breathing techniques on the male participants. The greater joint stiffness of men (13) may have contributed to lack of ventilatory effects. McHugh et al. (56) reported that mechanical factors contribute about 80 percent to the trunk forward flexion ROM, and thus the intrinsic male mechanical stiffness may have overcome neural or mechanical ventilatory effects. For both sexes in the study, EMG activity while inhaling before the stretch was not significantly lower, which tends to indicate that relaxed trunk muscle activity is not the single most important factor for attaining greater flexibility, contradicting some earlier studies (15, 64). Furthermore, pulmonary stretch reflexes inhibit sympathetic nerve activity at higher lung tidal volumes (71), thus there could have been lower sympathetic nerve activity for the pre-stretch inhale, inhale-during stretch, and hypoventilation conditions in the Hamilton study. Hamilton recommended that women should inhale at a slow frequency (hypoventilation) before and maintain that inhalation during the stretch. In contrast to the earlier studies cited above, they found that forceful exhalations actually increased EMG activity, thus inhibiting hip flexion ROM.

Combining Stretching with Muscular Contractions or Massage

Other techniques might be applied during a stretch to augment ROM increases. Implementing muscular contractions during a static stretch (active static stretch) might provide some greater benefit early in a training programme (i.e., four weeks) but no additional benefit over a longer time period

compared to passive stretching (i.e., eight weeks) (26). Another technique that can augment the stretching effect is massage of the tendon. In a study from our laboratory (37), friction massage was applied to the hamstrings tendon for either ten or 30 seconds prior to testing for hip flexion (hamstrings) ROM. Both durations provided 6–7 percent increases in ROM. The heat-induced friction of the massage would have decreased visco-elasticity and perhaps activated cutaneous and myofascial afferents helping to inhibit reflex-induced contractions. Massaging the tendon either before or during the stretch can augment the effectiveness.

Effect of Temperature on Stretching

Another consideration would be to ensure the tissues are at a higher temperature (hyperthermia) when stretching. Increased tissue temperatures decrease tissue stiffness and increase extensibility (48, 59, 62). Tendon temperatures over 39.4°C (103°F) can augment plastic elongation (48, 50), while tissue temperatures over 40°C (104°F) can enhance the viscous stress relaxation for collagen protein also leading to greater plastic deformation (54, 62, 63). There are conflicting studies regarding hyperthermic applications with increased ROM found following hot baths (72), hot packs (10 minutes) (51), ultrasound (83), and diathermy (60). But a lack of ROM augmentation has been reported after applying moist heat (70), or an electric heating pad (20 minutes at 43°C) (36).

There is controversy over whether a hypothermic (cold) stretch provides the best tissue lengthening (35, 51), or whether the hypothermic elongated position should be maintained until the tissue cools (e.g., application of ice) (49, 69). Cryotherapy (application of cold) in conjunction with static (58) and proprioceptive neuromuscular facilitation (PNF) (20) stretching has demonstrated improved flexibility compared to stretching alone. Cryotherapy can induce anaesthetic effects which would allow the individual to push past their usual point of discomfort and stretch farther (increased stretch tolerance). However, as with almost every scientific question, there are dissenting findings (19, 57, 66). For example, there were no significant improvements in ROM with the application of ten minutes of cold water immersion (14) or ice packs (51). Although, hypothermic applications (cryotherapy) can aid in pain tolerance, it would also lead to vasoconstriction (14) and increase tissue visco-elasticity (69). Sapega et al. (69) suggests that cold should be used for therapeutic conditions when the objective is to disrupt adhesions or there is substantial muscle spasticity.

Stretching under Metastable Conditions

There are some innovative stretch coaches who have combined stretching routines under relatively unstable conditions. For example, Mario DiSanto of Argentina (a former national gymnastics champion) has gymnasts place one foot on the floor with another foot suspended from a Theraband (elastic

band) attached to the ceiling. The athlete must stretch their muscles while maintaining stability and balance. A possible advantage of this type of active static stretching is the task specificity (11). Whereas, many stretching routines are performed under passive (no active voluntary muscles contraction) conditions, it is difficult to conceive of many sports that do not involve voluntary contractions and the necessity to maintain a high degree of balance or be in a state of strong metastability (39). Athletes and all individuals involved in activities of daily living continually move from states of stability (i.e., standing) to instability (movements such as walking and running) and return to stability again (i.e., return to the stance phase of walking or running). So, with a metastable state, an individual will move from a stable to a transiently unstable state (i.e., gymnast or figure skater or skier leaves the ground/floor to perform a jump or flip or other manoeuvre) and then return to their stable condition again. In many of these manoeuvres, a high degree of flexibility is necessary and the athlete certainly is not in a passive state. Stretching using unstable devices to create a metastable condition, accentuate task or action or sport specificity by placing the musculotendinous unit in a lengthened state under active metastable conditions. Thus, it is not only a stretch workout but a technique and motor control session as well (see Figure 7.1).

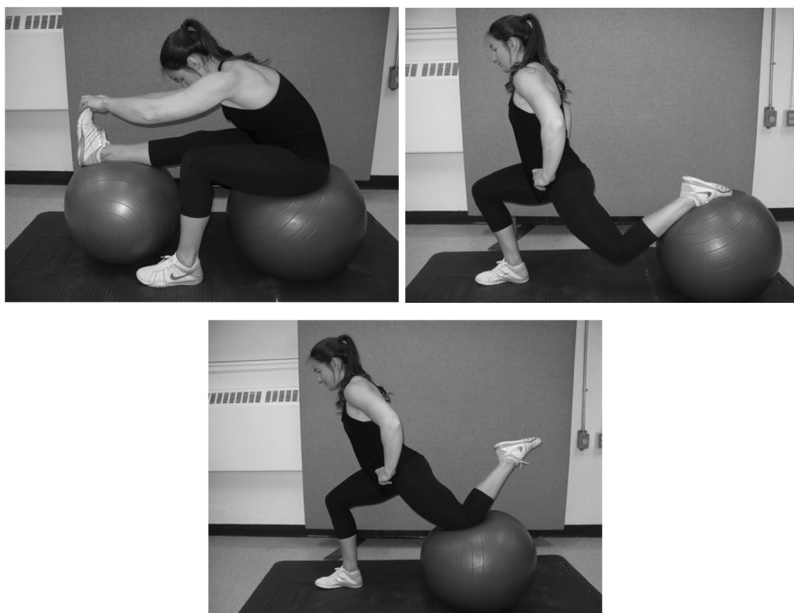


Figure 7.1 Stretching under unstable conditions to improve ROM and metastability

Summary

In summary, whereas single static stretches of five seconds can improve ROM, it is generally recommended that longer durations of 30–60 seconds provide optimal improvements in flexibility. It is not necessary to perform these stretches to the point of discomfort (100 percent intensity). While some studies have shown improvements even when stretching at 30–40 percent of maximal intensity, it seems that a stretching intensity of 60–85 percent would provide the greatest benefits. There are diurnal variations with flexibility and stretching midday to later in the day when the body is warmer decreases visco-elasticity and should enhance ROM. It is not recommended to stretch your back early in the morning when there is increased vertebral disc fluid pressure that could lead to disc protrusion injuries or nerve entrapment. There is no strong evidence for an optimal stretching frequency. However, since stretching is typically lower in intensity than resistance and high intensity anaerobic training without significant protein catabolism or tissue damage, it can be practised every day. Although, significant increases in flexibility are also experienced when stretching two or 3–six days per week.

The Konrad et al. meta-analysis (42) seems to be an irritating contrasting analysis since the lack of significant differences with their subgroup analyses of stretching volumes, intensities, and weekly frequencies contrasts with some of the original research articles. As I have told my students innumerable times: one study proves nothing! We need to look at the literature as a whole and that is the strength of meta-analyses. However meta-analyses are also not perfect. They attempt to congregate the findings from related but many times disparate groups and measures. For example, Konrad et al. (42) in their limitations section indicated that there was a moderate to high heterogeneity ($I^2 = 74.97$), owing to the wide variety of outcome measures, participants, and intervention durations. Thus, when you combine a wide distribution of findings, there will be higher variability and more difficulty in identifying anything significant. They also indicated the funnel plot as well as the significant Egger's regression intercept test revealed reporting bias. For example, significant positive results are more likely to be published in higher impact journals, producing greater citations numbers (24, 74). So, the good news is that it seems in general that many combinations of high or low stretching volumes, intensities or frequencies can be effective for improving ROM, whereas some original pieces of research suggest that for particular groups or scenarios there may be advantages with more strenuous or frequent stretching.

Whereas pre-event prolonged static stretching is not recommended, owing to the possibility of performance impairments, short to moderate duration static stretching (<60 seconds per /muscle group) within a full warm-up including dynamic stretching and dynamic activity does not impair performance (See Chapter 8). Post-activity stretching might involve a fatigued musculotendinous unit that could be susceptible to injury if subjected to

intense elongation. Thus, post-activity stretching should be lower intensity, which can promote not only improvements in ROM but psychological feelings of relaxation. When stretching, breathing patterns should be at a slow and controlled rate with a full tidal volume (large breaths) to promote greater relaxation. The breath should be expired or held as you approach the end of the ROM. While stretching a warm muscle decreases visco-elasticity promoting a greater ease of ROM, the use of cryotherapy especially with intense stretching or rehabilitation may increase stretch (pain) tolerance allowing the individual to push farther than normal.

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